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Practical Assessment - Halo IT - Data & AI Specialist

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Review

About Dataset -

The primary dataset was the support-ticket dataset supplied as part of the practical assessment. It contained ticket-level information for 2024–2025, including issue categories, priorities, status, assigned teams, response times, resolution times, SLA results, escalations and customer satisfaction.

I also incorporated New Zealand public-holiday data as the external data source. I chose this because holidays can affect staffing levels, ticket demand, response times and backlog. This made it possible to analyse whether operational performance changed before, during or after public holidays.

Process of working -

I first reviewed the business questions and identified the information an operations manager would need. I then inspected and cleaned the data, standardised categories and subcategories, corrected data types and created derived fields for open tickets, SLA results, ageing bands, holiday windows and day types.

After preparing the data, I created DAX measures for the key metrics, including total tickets, backlog, SLA compliance, response time, resolution time, escalation rate and CSAT. I then designed five report pages covering executive performance, issue analysis, SLA performance, team performance and holiday impact.

Finally, I reviewed the dashboards for usability, checked calculations, added filters and included operational insights to support decision-making.

Tools & programs used -

I used Python and Jupyter Notebook for data inspection, cleaning, validation and combining the ticket data with the external holiday data. Python was suitable because it allowed repeatable data-processing steps and made it easier to identify missing values, inconsistent categories and data-quality issues.

I used Power BI for reporting because it provides interactive slicers, DAX calculations, drill-down capabilities and multiple connected report pages. It is also appropriate for operational reporting because managers can quickly filter the dashboard by team, priority, category, region and time period.

I also created a Python-based AI assistant to answer natural-language questions about ticket trends, team performance and operational concerns.

Challenges & other alternative approach -

With more time, I would create a stronger star-schema data model with separate date, holiday, customer, team and issue dimension tables. I would also add automated data-quality tests and historical daily backlog snapshots.

The most challenging parts were defining consistent category mappings, handling different Boolean and date formats, creating meaningful ticket-ageing logic for a historical dataset and ensuring that SLA and team-performance measures were interpreted fairly.

Another challenge was presenting analyst performance with the correct context. Resolution time alone can be misleading because analysts may manage different levels of urgency and complexity.

Time spent on each phase (roughly) -

- Understanding the scenario and reviewing the data: approximately 1–2 hours
- Data cleaning, category mapping and external-data integration: approximately 3–4 hours
- Creating calculations and validating metrics: approximately 2–3 hours
- Building and formatting the Power BI dashboard: approximately 5–7 hours
- Developing and testing the AI assistant: approximately 3–4 hours
- Final validation, documentation and review: approximately 1–2 hours

Overall, I spent approximately **15–22 hours** on the practical assessment.

AI Tools -

Yes. I used AI tools as a development and review assistant. They helped me refine Python and DAX calculations, troubleshoot technical errors, evaluate dashboard layouts and improve documentation.

I also used an AI model as part of the practical solution by building a natural-language support-data assistant. The assistant can respond to questions about ticket trends, team performance and operational problem areas.

All calculations, outputs and recommendations were reviewed against the underlying dataset. AI was used to accelerate development and support problem-solving, rather than replace data validation or business judgement.

Summary -

This exercise was intended to assess my ability to clean and combine data, build an interactive operational dashboard, and use AI to support practical business decision-making. Through the solution, I demonstrated skills in Python, Power BI, DAX, external data integration, KPI design, data visualisation, and natural-language analysis. I was also able to show how technical outputs can be translated into clear operational insights and recommendations for management.